

Decentralized UAV-UGV Coordinated Navigation under Connectivity Constraints

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Abstract—This paper presents a novel decentralized navigation framework for urban environments, integrating an Unmanned Aerial Vehicle (UAV) with an Unmanned Ground Vehicle (UGV). The system is designed such that the UAV aims to reach a predefined destination to perform critical aerial tasks while the UGV maintains proximity to support dynamic charging and continuous operation. The navigation strategy employs a hybrid path planning approach, combining a Probabilistic Roadmap (PRM) with an A* search algorithm, further optimized by path pruning for the UAV. In contrast, the UGV utilizes the Dynamic Window Approach (DWA) for adaptive navigation. Both unmanned vehicles operate while maintaining a safe distance to ensure uninterrupted communication and prevent connectivity loss. Validated through extensive simulations in ROS2 and Gazebo, this framework enhances mission efficiency and enables real-time coordination for effective decentralized navigation.

Index Terms—UAV, UGV, autonomous navigation, coordination, path planning.

I. INTRODUCTION

Over the past decades, unmanned systems have experienced remarkable advancements driven by improvements in computation, sensing, and control technologies. Unmanned Aerial Vehicles (UAVs) have emerged as agile platforms capable of rapid deployment and aerial reconnaissance, while Unmanned Ground Vehicles (UGVs) offer robust endurance and higher payload capacities. Despite their individual strengths, standalone UAVs and UGVs are often constrained by limitations such as restricted energy capacity, mobility challenges, and limited operational range. Recent trends in robotics research have increasingly focused on integrating heterogeneous platforms into cooperative systems, a development that unlocks new possibilities for tackling complex operational challenges in dynamic, obstacle-rich environments. Applications of UAV-UGV systems are diverse and include precision agriculture, surveillance, and infrastructure inspection to search and rescue operations in disaster-affected areas and environmental monitoring of hazardous terrains [1]–[4].

Recent research on integrated UAV-UGV systems demonstrates a broad range of applications and promising advancements in cooperative robotic platforms. Qin et al. [5] introduced a low-cost aerial-ground delivery system that harnesses the complementary capabilities of UAVs and UGVs, emphasizing cost efficiency and operational flexibility in logistics. Similarly, Singh et al. [6] explored UAV and UGV-assisted path planning [7] for sensor data collection in precision agriculture, showcasing the potential to enhance data acquisition and crop monitoring. In [8], the authors investigated cooperative 3D multi-object tracking by integrating multi-source data fusion, significantly improving situational awareness and tracking accuracy through the joint perspectives of aerial and ground vehicles. Parallel to these developments, efforts have

been made to address the coordination and resource allocation challenges inherent in UAV-UGV systems. Yan et al. [9] examined formation tracking using event-triggered communications, ensuring robust and efficient coordination between platforms under communication constraints. Moreover, Ma et al. [10] proposed an adaptive depth graph neural network framework for dynamic task allocation in complex environments, enhancing the system's adaptability and operational robustness. Together, these studies highlight the evolving capabilities of integrated UAV-UGV systems and underscore a trend toward more efficient, resilient, and versatile multi-agent robotic platforms in both civilian and defense applications. Despite these promising contributions, several limitations persist in current UAV-UGV cooperative systems. Many studies, while innovative in their respective application areas, have essentially focused on the application aspect of the UAV-UGV system, developed centralized navigation approaches, or not considered real-time coordination under communication limitations.

Building on the insights discussed, this paper proposes a novel coordination UAV-UGV navigation framework in obstacle-constrained environments where the UAV performs mission-critical tasks and aiming at reaching pre-defined destinations, while the UGV functions as a mobile unit aiming at following the UAV, e.g., mobile charging station. First, we propose a hybrid global path planning strategy that combines a Probabilistic Roadmap (PRM) with an A* search algorithm, incorporating a path pruning mechanism to optimize the planning process of the UAV. Second, we adopt the Dynamic Window Approach (DWA) to continuously reposition the UGV to stay in close proximity to the UAV. The coordinated navigation is performed in a decentralized manner while avoiding connectivity loss between the unmanned vehicles. The framework is validated through simulations in ROS2 and Gazebo, demonstrating effective real-time coordinated navigation. This approach diverges from traditional methods that often handle aerial and ground vehicles separately or rely on reinforcement learning solutions, which may compromise accuracy and safety [11], [12].

II. SYSTEM MODEL

This section outlines the modeling for the UAV-UGV system. It also introduces the environmental representation and the mathematical models for both vehicles. As shown in Figure 1, the UAV navigates complex, obstacle-rich environments while coordinating its movement with the UGV.

A. Mapping and Environment Representation

We consider that unmanned vehicles are provided with a 3D map of the environment in which they are navigating. This

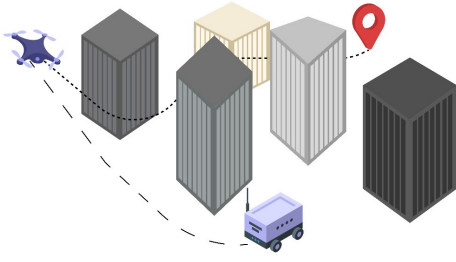


Fig. 1: Illustration of the UAV navigation in an urban environment.

raw map is first converted into a structured 3D Point Cloud Data (PCD) map, which is then processed to extract relevant information such as free space, obstacles, and navigable areas. This processed data is used by the UAV for global path planning and by the UGV for local navigation, as it will be detailed in Section III. The environment is modeled as a multi-dimensional navigation space, typically using:

1) *Grid Representation:*

a) *For UAV Navigation:* The UAV operates in a bounded volumetric space defined as $[X_{UAV}, Y_{UAV}, Z_{UAV}]$, which is discretized into voxels or occupancy grids with resolution-dependent cells for path planning. Thus, the number of voxels N_v can be calculated according to a chosen resolution ρ as follows:

$$N_v = \left(\frac{X_{UAV}}{\rho}\right) \times \left(\frac{Y_{UAV}}{\rho}\right) \times \left(\frac{Z_{UAV}}{\rho}\right). \quad (1)$$

b) *For UGV Navigation:* The UAV moves on a planar surface defined as $[X_{UGV}, Y_{UGV}]$, which is discretized into a grid of cells according to ρ , resulting in a number of cells N_c measured as follows:

$$N_c = \left(\frac{X_{UGV}}{\rho}\right) \times \left(\frac{Y_{UGV}}{\rho}\right). \quad (2)$$

2) *Static Obstacles:* In this study, we consider static objects placed on the map, such as buildings, walls, and trees. They are represented as occupancy grids, where each cell (or voxel) is marked as free or occupied, forming an occupancy map $M(x, y, z)$ defined as follows:

$$M(x, y, z) = \begin{cases} 1, & \text{if occupied,} \\ 0, & \text{if free.} \end{cases}$$

For map generation, the system can utilize pre-recorded offline maps to create a 3D representation of the environment. The processed PCD map is fed into the UAV's global planner to compute a collision-free path, while the UGV integrates this data into the Navigation2 (Nav2) stack, allowing it to reactively follow the UAV while avoiding obstacles. This unified mapping framework ensures that both unmanned vehicles operate in a synchronized and environment-aware manner.

B. UAV Model

The UAV, identified as a quadrotor system and characterized by the position vector $\mathbf{r}$, representing the center of mass in the world coordinate frame, is tied to the UAV's dynamics, as expressed in the following [13]:

$$m\ddot{\mathbf{r}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + R \begin{bmatrix} 0 \\ 0 \\ F_1 + F_2 + F_3 + F_4 \end{bmatrix}, \quad (1)$$

$$I \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} L(F_2 - F_4) \\ L(F_3 - F_1) \\ M_1 - M_2 + M_3 - M_4 \end{bmatrix} - \begin{bmatrix} p \\ q \\ r \end{bmatrix} \times I \begin{bmatrix} p \\ q \\ r \end{bmatrix}, \quad (2)$$

where F_1, F_2, F_3 , and F_4 represent the thrust forces generated by the four individual rotors of the quadrotor UAV, and M_1, M_2, M_3 , and M_4 denote the moments or torques generated by the rotors. The parameter m denotes the UAV's mass, and R is the rotation matrix that converts the body frame into the world frame using the $Z-X-Y$ Euler angle convention. The symbol I represents the moment of inertia about the UAV's center of mass, and L is the distance from the center of mass to each rotor's axis of rotation. The variables p, q , and r are the components of the angular velocity. The control inputs include two parts: the total thrust $u_1 = F_1 + F_2 + F_3 + F_4$ and the moments u_2 given as follows:

$$u_2 = \begin{bmatrix} L(F_2 - F_4) \\ L(F_3 - F_1) \\ M_1 - M_2 + M_3 - M_4 \end{bmatrix}. \quad (3)$$

The state vector of the UAV is defined as:

$$\mathbf{x}_{UAV} = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, p, q, r]^T \quad (4)$$

where $\mathbf{r} = [x, y, z]^T$, ψ, ϕ , and θ are the position, the yaw, the roll (rotation around x-axis), and the pitch (rotation around y-axis), respectively.

C. UGV Model

We assume that the UGV operates under non-holonomic constraints, meaning it cannot move sideways like a UAV or a holonomic robot (e.g., omnidirectional wheels). It can only move forward, backward, or rotate based on its wheeled configuration. The generalized state vector for a ground robot is given as follows:

$$\mathbf{x}_{GR} = (x, y, \theta, v, \omega)^T, \quad (5)$$

where (x, y) is the UGV's position in a 2D coordinate frame, θ is UAV's heading (yaw angle), v is the linear velocity, and ω is the angular velocity. The kinematic model of a ground robot follows:

$$\dot{x} = v \cos(\theta), \quad \dot{y} = v \sin(\theta), \quad \dot{\theta} = \omega, \quad (6)$$

where v is the linear velocity along the forward direction, and ω is the angular velocity determining the turning rate. In a Differential Drive Robot (DDR), the wheel velocities determine motion:

$$v = \frac{r_w}{2} (\omega_L + \omega_R), \quad \omega = \frac{r_w}{d} (\omega_R - \omega_L) \quad (4)$$

where r_w is the wheel radius, d is the wheel separation distance, and ω_L and ω_R are the left and right wheel angular velocities.

D. UAV-UGV Channel Model

In this study, UAV and UGV communicate via an Air-to-ground (A2G) channel, exchanging positional data essential for the UGV to track the faster-moving UAV, which in turn must maintain a safe distance to avoid connectivity loss. We model the channel gain H as a function of distance d , considering the large-scale path loss: $H(d) = \frac{1}{\sqrt{PL(d)}}$, where $PL(d)$ represents the path loss, which varies with

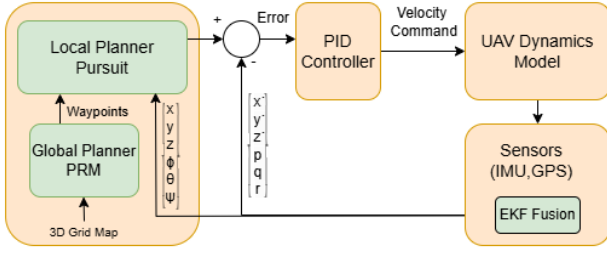


Fig. 2: UAV control flow diagram.

the Euclidean distance between the units. The path loss in decibels (dB) for line-of-sight (LoS) and non-line-of-sight (NLoS) conditions is modeled as [14]:

$$PL_{dB}(d) = p^{LoS} PL_{dB}^{LoS} + (1 - p^{LoS}) PL_{dB}^{NLoS}, \quad (5)$$

with the path loss for each condition given by:

$$PL_{dB}^X(d) = 10n \log_{10} \left(\frac{4\pi f d}{C} \right) + L_{dB}^X, \quad (6)$$

where $X \in \{LoS, NLoS\}$, f is the carrier frequency, n is the path loss exponent, C is the speed of light, and L_{dB}^X is the additional loss specific to each link condition. The probability of maintaining a LoS link is given as:

$$p^{LoS}(d) = \frac{1}{1 + \beta_1 \exp(-\beta_2[\theta(d) - \beta_1])}, \quad (7)$$

where β_1 and β_2 are parameters dependent on the environmental context, and $\theta(d)$ is the elevation angle between the UAV and the UGV.

To ensure uninterrupted connectivity, the path loss should be maintained at a sufficiently low level, such that the distance separating the UAV and UGV does not exceed ηD_{th} , where D_{th} is the threshold for safe distance and η is a tolerance parameter. This condition helps to avoid potential connectivity losses and ensures that both units operate within a safe operational range.

III. UAV-UGV COORDINATED NAVIGATION APPROACH

This section presents the main methods enabling UAV-UGV coordinated navigation in the obstacle-constrained environment where the UAV aims to reach a target location without losing connectivity with the UGV. The latter aims to keep close proximity with the flying unit.

A. UAV Navigation Framework

1) *Global Path Planning for UAV*: The shortest path on the graph isn't always the one on the environment because the PRM can be sparse. [15] As a result, we include a supplementary path simplification process to further optimize the produced path. By determining if direct connections between intermediate waypoints are feasible without running into barriers, this stage assesses whether the originally calculated path can be reduced. A more direct route eliminates pointless detours, cutting down on total trip distance and increasing efficiency.

With the help of this two stage method PRM for roadmap creation, A* for pathfinding, and a simplification check for optimization the UAV is guaranteed to take the most effective path while avoiding obstacles. PRM path is generated based on N number of nodes in the 3d space. The Pure Pursuit Path

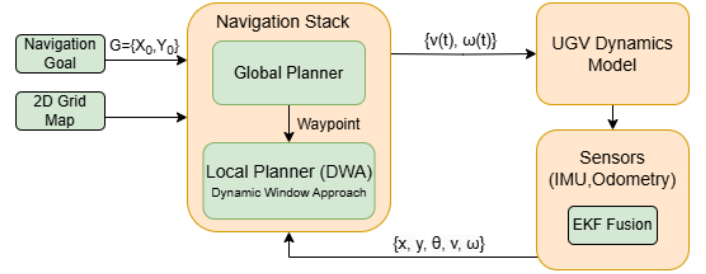


Fig. 3: UGV control flow diagram.

Solver then creates smooth trajectory commands for the UAV to follow, publishing the optimum path for execution. Fig. 3 depicts a thorough flow.

2) *Path Following and Control for UAV*: After the global path has been determined, the UAV follows the computed trajectory using a Pure Pursuit Path Solver [16]. This algorithm calculates a series of target waypoints along the planned path and continuously adjusts the UAV's heading and velocity to maintain smooth navigation. The Pure Pursuit method ensures that the UAV maintains a stable path, avoiding sharp turns or erratic movements. To further enhance trajectory tracking, a PID Controller is integrated to regulate velocity and orientation. The PID controller plays a pivotal role in our UAV control architecture by using continuous sensor feedback to adjust the control inputs and maintain the desired flight path. Specifically, the controller modulates the throttle, roll, pitch, and yaw based on the UAV's state, which is comprehensively defined by the state vector in Equation (4). This state vector provides detailed information about the vehicle's position, orientation, and velocity, which is essential for the PID controller to accurately compute the required adjustments. The final control outputs from the PID controller are then applied to the UAV's dynamics model, executing the planned motion. This multi-stage control framework enables precise and adaptive UAV navigation, making it robust for real-world deployment scenarios. The complete workflow for the UAV navigation is depicted in Figure 2.

B. UGV Navigation Framework

Unlike the UAV, which follows a predefined global path, we propose that the UGV dynamically adjusts its trajectory using the NAV2 stack and the DWA planner [17]. This choice allows the UGV to autonomously move while tracking the UAV. In fact, the NAV2 stack provides a flexible and modular navigation system that reacts to environmental changes. The DWA planner, used within NAV2, is responsible for local path planning and obstacle avoidance. DWA operates by evaluating possible velocity commands within a defined window, selecting the most optimal movement that avoids obstacles while keeping the UGV dynamically following the UAV without requiring a strict predefined path. The UGV navigation process follows these steps:

- **Map Processing:** The processed 3D PCD map is provided as input to NAV2, enabling the UGV to understand the navigable space.
- **Localization:** The UGV continuously tracks its position using sensor fusion (e.g., odometry and IMU).

Algorithm 1 UAV-UGV Coordination Framework

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1: Input: 3D map, starting UAV and UGV positions  $S, P$ ,
   UAV destination position  $G$ , number of PRM nodes  $N$ ,
   threshold distance  $D_{th}$ 
2: Output: Control commands for UAV and UGV
   ▷ UAV Path Planning:
3: Generate PRM with  $N$  collision-free samples
4: Compute optimal path  $P$  using A* and path pruning
   ▷ UAV-UGV Trajectory Execution (e.g., 10 Hz update
   rate)
5: for each control cycle do
   ▷ UAV Control:
6:   Calculate the look-ahead point on the path
7:   Compute and apply velocities and steering com-
   mands  $u_1$  and  $u_2$ 
   ▷ UGV Control:
8:   Receive UAV position update
9:   For velocity candidates  $(v, \omega)$  in (??), compute and
   execute optimal velocity command  $(v^*, \omega^*)$  to follow
   UAV using:
10:   $G(v^*, \omega^*) = \alpha \cdot \text{heading} + \beta \cdot \text{clearance} + \gamma \cdot \text{velocity}$ 
   ▷ Coordination Navigation:
11:  Exchange position data between UAV and UGV
12:  Compute current distance  $d$  between UAV and UGV
13:  if  $d > \eta D_{th}$  then
14:    Decrease UAV velocity to avoid connectivity loss
15:  elseif  $d \leq \eta D_{th}$  then
16:    Maintain command parameters
17:  endif
18:  Adjust UGV trajectory to maintain proximity
19: end for

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- **Path Planning:** DWA evaluates velocity and trajectory options, ensuring that the UGV moves safely while staying in proximity to the UAV.
- **Obstacle Avoidance:** The NAV2 stack automatically utilizes 3D map to identify obstacles and recalculating the optimal trajectory as needed.

C. Coordinated UAV-UGV Navigation

Coordination between the UAV and UGV is established through communication mechanisms, ensuring real-time data exchange for synchronized movement, as detailed in Algorithm 1. The UAV continuously publishes its position and planned path via ROS2 topics, which are then received by UGV to dynamically adjust the movement. The communication process includes:

- **Position Sharing:** The UAV broadcasts its current location and planned trajectory as shown in lines 14-15 of Algorithm 1.
- **Velocity Commands:** The UAV can send suggested velocity updates for smoother coordination, following the approach outlined in Algorithm 1.
- **Feedback Loop:** The UGV sends periodic status updates, allowing for adjustments in movement synchronization as per the coordination step in Algorithm 1.

Table I: Simulation Parameters

Parameter	Value	Parameter	Value
Map Size	40x40x10	Resolution	0.2m/voxel
Number of Voxels	2,000,000	Grid Dimensions	200x200x50
Cells (2D)	40,000	Cells (3D)	2,000,000
Robots Used	UAV, UGV	PRM Samples	2000 - 60000
K-Neighbors	10	Max Cost Value	100

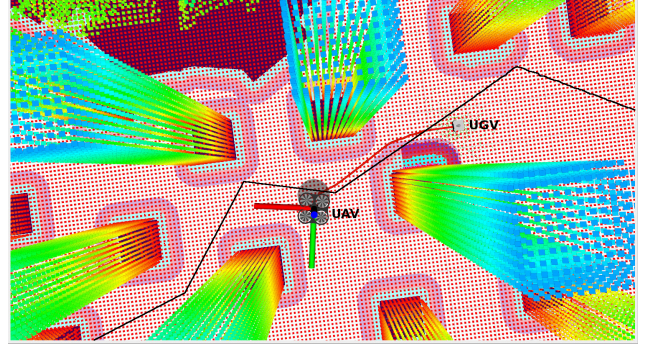


Fig. 4: Visualization of UAV-UGV coordination in a simulated environment. The UAV follows the A* with path pruning trajectory (black line), while the UGV dynamically adjusts its path (red line).

This communication framework enables the UGV to reactively follow the UAV, respecting the safe distance D_{th} computed from Section II-D, while navigating obstacles. The use of ROS2 for decentralized coordination ensures that both unmanned vehicles can operate *independently* while staying in sync, making the system scalable for future enhancements such as multi-UAV and multi-UGV coordination, building upon the framework described in Algorithm 1.

IV. SIMULATION RESULTS

This section presents the experimental results obtained from the UAV-UGV coordination system simulations on ROS2 and Gazebo environments. We evaluate the system's performance for the simulation parameters given in Table I.

Figure 4 illustrates the ROS2 and Gazebo simulation environment, providing a visual representation of the UAV and UGV coordination within a simulated urban scenario. The simulation demonstrates the UAV-UGV's ability to avoid static obstacles around the complex urban environment while maintaining inter-vehicle communication links to exchange data with the ROS2 communication framework.

Figure 5 compares the achieved travel time (left y-axis) and the execution time (right y-axis) of the proposed approach. i.e., PRM, standard A* algorithm, and the proposed A* with Path Pruning method for different PRM sample sizes ranging from 500 to 60,000 nodes. It compares the performance to the case where PRM and A* are executed without the path pruning method. It is shown that the A* with Path Pruning method achieves an approximate 33% reduction in travel time compared to standard A*, highlighting its efficiency in determining more effective path for the UAV.

On the other hand, the figure displays the time required to generate the PRM given the needed number of nodes N . It is shown that the computational complexity is high (i.e., the order of seconds) for sufficiently large PRM and trivially increases with the number of sampled nodes. However, it is

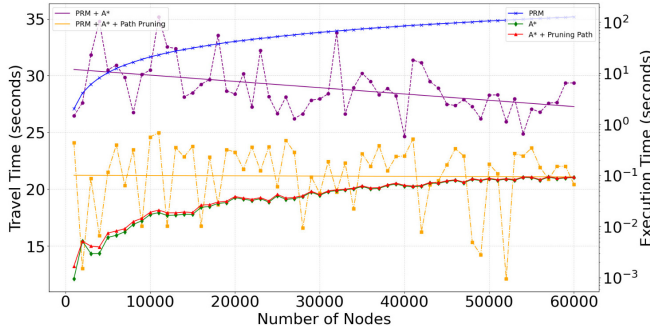


Fig. 5: Performance comparison of the UAV path planning algorithms: travel time (right y-axis) and algorithm execution time (left y-axis) versus different PRM nodes (N).

Table II: Average mission time, energy consumption, and distance for UAV+UGV vs. standalone UAV

Vehicle	Avg. Time (s)	Avg. Energy (J)	Avg. Distance to Charging Station (m)
UAV + UGV	41.61	6383	5.57
UAV	22.88	3498	26.9

worth to note that the PRM is generated once per scene and there is no need to execute to determine the optimal path for another UAV stating position and destination. On the other hand, we can notice that the time execution of the A* (with and without path pruning) is relatively low (order of tens of milliseconds), which is convenient for real-time navigation. Finally, it is noteworthy that the path pruning step enabling significant saving in terms of travel time requires a negligible computational overhead.

Finally, in Figure 5, we notice that, on average, the travel time is constant with A* with path pruning method with respect to the PRM size unlike the case with the standalone A*, which allows obtaining effective path for lower PRM size on average, and hence, considerable computational saving.

Table II summarizes the average mission time, energy consumption, and distance to the charging station for the flying unit for two configurations operating for the same trajectory in the same environment: a standalone UAV, assuming fixed charging station located at the stating point, and a UAV-UGV pair where the UGV acts as a mobile charging station. It is shown that the UAV+UGV setup leads to a longer mission time and higher energy consumption compared to a standalone UAV (stationary charging station). Technically, this arises because the UAV must hover whenever the UGV lags behind to avoid connectivity loss causing additional power draw and time overhead. However, despite the increased time and energy, the UAV+UGV system enables the UAV to maintain close distance to the charging station indicating a reduced need for long return trips, which increases the operation range of the UAV compared to the traditional scenario.

A stationary charging station would require the UAV to periodically detour back to a single, fixed location for recharging, potentially resulting in even larger travel distances and longer mission interruptions. In contrast, the movable station follows the UAV throughout the environment, providing a localized recharge option that can be accessed with less detour. This trade-off becomes especially relevant in large-scale or time-critical missions, where minimizing the UAV's return-to-base cycles can outweigh the additional hover overhead.

V. CONCLUSION

This paper presented a decentralized navigation framework for UAV-UGV systems under connectivity and urban operational constraints. By integrating a hybrid path planning strategy with dynamic coordination mechanisms, the framework ensures robust decentralized UAV-UGV coordination. Simulations validated the effectiveness of the proposed approach, demonstrating significant savings in execution time and mission efficiency. Future work will focus on testing the framework for real-world unmanned vehicles and validating their practicality and adaptability in diverse environmental scenarios.

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